

Where the switch cost lives in active inference

Notebook 11’s attention rule, decomposed against the EFE formalism

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Abstract

Notebook 11’s deterministic attention rule “switch iff $\text{EIG}_{\text{best}} > \text{EIG}_{\text{stay}} + c$ ” was introduced as an attentional *switch cost*. A forest-walk discussion asserted that this corresponds to the *pragmatic* term of expected free energy. **That assertion is wrong.** The pragmatic term is a preference over *outcomes*; the switch cost is a cost on *actions*. This note shows, against the equations of Da Costa et al. (2020) and Friston et al. (*Active inference and learning*, 2016), that the switch cost is instead a *policy prior* $E(\pi)$ — the habit / control-cost slot of the policy posterior $\sigma(\ln E - \gamma G)$. Notebook 11’s hysteresis rule is precisely the MAP / winner-take-all action selected from that posterior with flat outcome preferences and a persistence-favouring habit. The mechanism is therefore principled in *type*; whether the specific functional form of $\ln E$ is independently anchored (e.g. in attentional dwell-time data) is a separate, residual question.

1 Lead

The forest-walk claim was: *the switch cost is the pragmatic term of EFE that the EIG-only policy was missing*. Reading Da Costa et al. (2020) carefully, this is wrong. The pragmatic term of $G(\pi)$ is $-\mathbb{E}_{Q(o_\tau|\pi)}[\ln P(o_\tau)]$ — an expectation over *outcomes* o_τ , weighted by the preference prior $C \equiv P(o_\tau)$. A switch cost is attached to an *action*; it is not in the outcome slot.

The honest correction: the switch cost lives in $\ln E(\pi)$, the *policy prior* (Da Costa footnote 5; Friston *Active inference and learning*, Eq. 7). Equivalently, in the optimal-control idiom, it is a *KL-control cost* for deviating from a default “stay” policy. The two pictures in this note — the anatomy of the policy posterior (Fig. 1) and the hysteresis decision rule (Fig. 2) — pin both the location of the slot and the exact form of the rule.

2 The expected free energy and where each piece lives

Da Costa et al. (2020) define the policy posterior (their Eq. 10) as

$$Q(\pi) = \sigma(-G(\pi)), \quad (1)$$

with the expected free energy admitting two decompositions of interest. The *risk + ambiguity* form (their Eq. 13):

$$G(\pi) = \underbrace{D_{KL}[Q(s_\tau, A | \pi) \parallel P(s_\tau, A)]}_{\text{Risk}} + \underbrace{\mathbb{E}_{Q(s_\tau, A|\pi)}[H[P(o_\tau | s_\tau, A)]]}_{\text{Ambiguity}}. \quad (2)$$

And the *value* decomposition (their Eq. 16):

$$\begin{aligned}
G(\pi) = & \underbrace{-\mathbb{E}_{Q(o_\tau|\pi)}[\ln P(o_\tau)]}_{\text{Extrinsic / pragmatic}} \\
& - \underbrace{\mathbb{E}_{Q(o_\tau|\pi)}[D_{KL}[Q(s_\tau | o_\tau, \pi) \| Q(s_\tau | \pi)]]}_{\text{Saliency (info gain about states)}} \\
& - \underbrace{\mathbb{E}_{Q(o_\tau, s_\tau|\pi)}[D_{KL}[Q(A | o_\tau, s_\tau, \pi) \| Q(A)]]}_{\text{Novelty (info gain about parameters)}} - (\text{evidence bound}). \tag{3}
\end{aligned}$$

Two readings off Eq. (3).

The pragmatic term is over outcomes. $-\mathbb{E}_{Q(o_\tau|\pi)}[\ln P(o_\tau)]$ is an expectation under the predicted-outcome distribution, weighted by $C \equiv P(o_\tau)$. The review text is unambiguous: “higher-valued *outcomes* have higher prior probability under $P(o)$ reflecting agent preferences — **not costs on actions themselves.**”

Notebook 11’s EIG is the novelty term. $\mathbb{E}[D_{KL}[Q(A | o, s, \pi) \| Q(A)]]$ is the expected information gain about a model *parameter*. The closed-form $\frac{1}{2} \log \det(\Lambda + hh^\top / \sigma^2) - \frac{1}{2} \log \det(\Lambda)$ implemented by `expected_info_gain_vec` is exactly that quantity for the linear-Gaussian model over θ . Da Costa: “*It is this term that underwrites artificial curiosity . . . maximising intrinsic value corresponds to optimal Bayesian design.*” So one half of nb11’s policy is principled and correctly named.

3 Anatomy of the policy posterior

The full policy posterior, with the prior over policies made explicit, is given by Friston et al. (*Active inference and learning*, Eq. 7):

$$Q(\pi) = \sigma(\ln E(\pi) - \gamma G(\pi)). \tag{4}$$

Da Costa’s footnote 5 confirms the same structure: “*A more complete treatment may include priors over policies — usually denoted by E . . . leading to an approximate posterior of the form $\sigma(-\ln E - F - G)$.*” Friston: “ *$E(\pi)$. . . creates an implicit cost structure through prior preferences rather than explicit penalties,*” with a designated *habitual* policy that the prior can favour. This is the slot a cost-for-deviating-from-a-default-policy lives in.

Figure 1 pins each of nb11’s pieces to its location in that anatomy.

4 Why the switch cost is not the pragmatic term

A switch cost is attached to an *action* (switching attention). The pragmatic term is an expectation over *outcomes* (Eq. (3)). There is no construal under the standard discrete-state EFE in which “penalise the switch action” becomes $-\mathbb{E}[\ln P(o_\tau)]$. The forest-walk identification was a type-error.

5 Where it actually lives: the policy prior $E(\pi)$

The policy posterior (4) contains an additive $\ln E(\pi)$ term that does not depend on $G(\pi)$ at all. Setting

$$\ln E(a) = -c \cdot \mathbf{1}[a \neq a_{\text{prev}}] \tag{5}$$

is a log-policy-prior that favours the persistence / habitual policy by a fixed log-odds ratio c relative to any deviation. This is the standard “habit” form (Friston, *Active inference and learning*, §2).

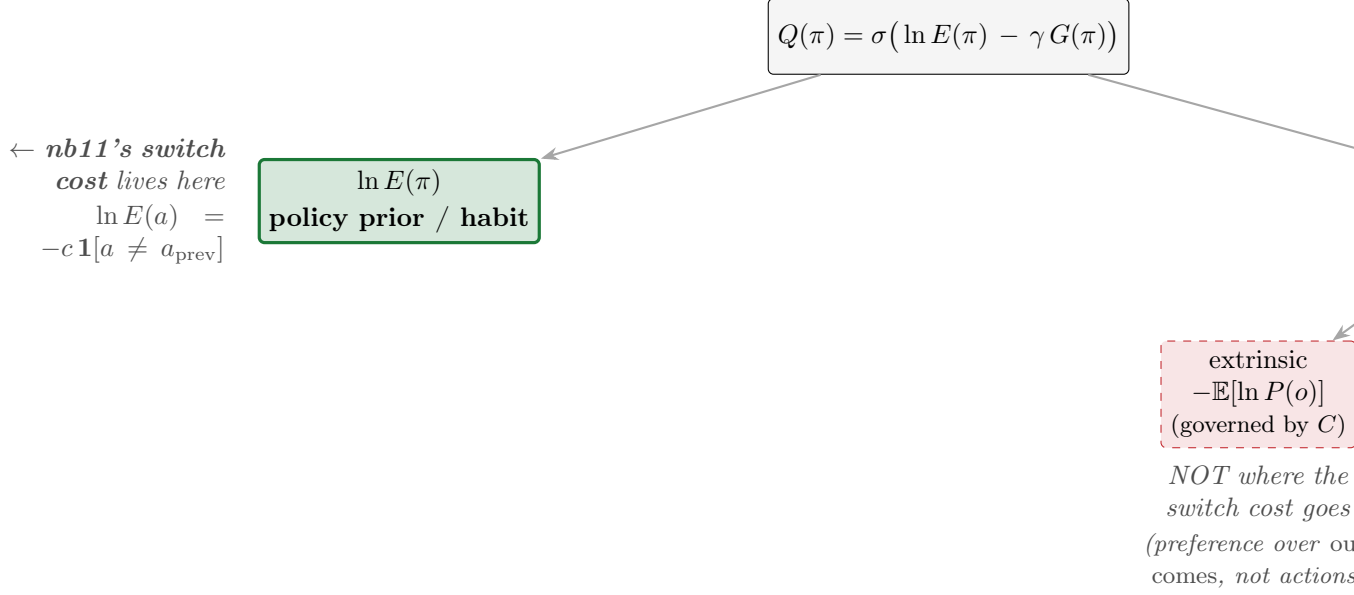


Figure 1: Anatomy of the active-inference policy posterior, with notebook 11’s two ingredients pinned to their slots. The switch cost sits in the **policy prior** $E(\pi)$ (green, top-left), *not* in the extrinsic / pragmatic value (red, bottom-left). The EIG is the **novelty** term of $-G(\pi)$ (green, bottom-right).

6 Equivalence: nb11’s rule is MAP under the policy posterior

Notebook 11’s deterministic attention rule is, with c for the switch cost,

$$a_t = \begin{cases} \arg \max_a \text{EIG}_a & \text{if } \max_a \text{EIG}_a > \text{EIG}_{a_{\text{prev}}} + c, \\ a_{\text{prev}} & \text{otherwise.} \end{cases}$$

Proposition 1. *Let $V(a) := \text{EIG}_a - c \mathbf{1}[a \neq a_{\text{prev}}]$ with $c > 0$. Then $a_t = \arg \max_a V(a)$, and equivalently a_t is the MAP action under the active-inference policy posterior (4) with novelty EIG_a , flat outcome preferences (extrinsic value constant in a , so $-\gamma G(a) = \gamma \text{EIG}_a + \text{const}$), EFE precision $\gamma = 1$, and policy prior (5). In the high-precision limit (Da Costa §6.3), the softmax collapses to argmax, so the deterministic rule is exactly this MAP.*

Proof. Let $M = \max_{a \neq a_{\text{prev}}} \text{EIG}_a$ and $S = \text{EIG}_{a_{\text{prev}}}$. From (5), $V(a_{\text{prev}}) = S$ and $V(a) = \text{EIG}_a - c$ for $a \neq a_{\text{prev}}$. Thus $\arg \max_a V(a) = a_{\text{prev}}$ iff $S \geq M - c$, i.e. iff $M \leq S + c$; otherwise $\arg \max_a V(a) = \arg \max_{a \neq a_{\text{prev}}} \text{EIG}_a$.

Now compare to nb11’s rule. Let $B = \max_a \text{EIG}_a$. *Case A:* the global argmax is a_{prev} , so $B = S$. Then $B > S + c$ is false ($c > 0$), so nb11 stays. And $M \leq S \leq S + c$, so V -argmax also stays. *Case B:* the global argmax is some $a^* \neq a_{\text{prev}}$, so $B = M$. Both rules switch iff $M > S + c$, and both then go to $\arg \max_{a \neq a_{\text{prev}}} \text{EIG}_a = a^*$. $\square$

Figure 2 visualises the rule on three bands.

7 The KL-control reading

Da Costa’s text under Eq. (2): “In the optimal control literature, this form of expected free energy underwrites **KL control**” (Todorov 2009; Kappen, Gómez & Opper 2012). KL control parametrises the cost of a controlled policy as a KL divergence from a *passive / default* policy — mathematically the same slot as $\ln E(\pi)$. Notebook 11’s $\mathbf{1}[a \neq a_{\text{prev}}]$ with default “stay” is

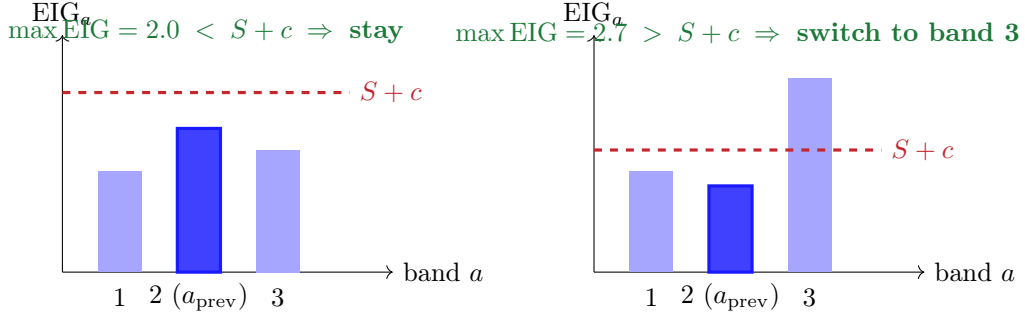


Figure 2: The hysteresis rule visualised. Bars show the three bands’ EIG_a ; the highlighted bar is the current band (a_{prev} , with value S); the red dashed line is the switching threshold $S + c$. Equivalently, the agent picks $\arg \max_a [\text{EIG}_a - c \mathbf{1}[a \neq a_{\text{prev}}]]$ — the persistence default gets a free pass; every other option pays a tax of c .

exactly a discrete, hard-threshold control cost on deviation from the passive policy. Whether one names it “policy prior,” “habit,” or “KL-control cost” is a matter of register; the formal object is one and the same.

8 Honest bottom line

- **Principled in *type* — yes.** The switch cost populates a standard, named slot in the active-inference policy posterior: the policy prior $E(\pi)$ / habit / KL-control cost. It sits additively beside $-\gamma G(\pi)$. nb11’s hysteresis rule is exactly the MAP action under that posterior with flat preferences and a persistence-favouring habit (Proposition 1). The forest-walk worry that the mechanism is *ad hoc* machinery is not borne out: the EIG-only policy was simply missing the $E(\pi)$ term, and the switch cost is the (MAP limit of) it.
- **Principled in *form and value* — not automatically.** The formalism *permits* a policy prior; it does not *dictate* the flat additive form $\ln E(a) = -c \mathbf{1}[a \neq a_{\text{prev}}]$ or any particular value of c . Choosing the simplest possible form, and tuning c until the network oscillated, remain researcher degrees of freedom. The framework legitimises the *slot*; it does not absolve the *choice*. The straightforward fix: anchor c in attentional dwell-time / task-switching-cost literature (Monsell; Duncan, Ward & Shapiro), so what was a free parameter becomes either a prediction or a measurement-driven constraint.
- **One footnote, for completeness.** In the strong active-inference claim that “there are no cost functions, only priors,” action / effort costs get reified as proprioceptive *outcome* preferences and folded back into C . Under that reading the forest walk could be partially salvaged — but the standard discrete-state formulation (Da Costa) puts the cost in $E(\pi)$, and that is the cleaner home for nb11.

For nb11. Recast the parameter from `switch_cost` to “persistence policy prior” with the equation $\ln E(a) = -c \mathbf{1}[a \neq a_{\text{prev}}]$, cite Da Costa footnote 5 and Friston *Active inference and learning* for the slot, cite the KL-control lineage for the form, and state plainly that the form is the simplest choice and the value should ideally be anchored to dwell-time data. This is both correct and a stronger framing than “we added a switch cost.”

References

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